

ChromaDB Extraction Pipeline Architecture

1. Overview

This document outlines the extraction pipeline (RAG architecture) used to pull regulatory rules from the local ChromaDB vector store. The database stores parsed RBI regulations, Anti-Money Laundering (AML) policies, and suspicious activity definitions.

2. Pipeline Components

Component	Technology	Description
Embeddings	Ollama (nomic-embed-text)	Generates highly semantic 768-dimensional vectors from input text
Vector Store	ChromaDB	Local vector database running on port 8001 via Docker, storing embeddings
Integration	LangChain	Orchestrates the connection between the embedding model, ChromaDB, and the LLM
Similarity Metric	L2 (Euclidean) / Cosine	Calculates the mathematical distance between the transaction query and document embeddings

3. Step-by-Step Extraction Flow

Step 1: Context Preparation

The Compliance Agent detects a suspicious transaction and generates a natural language query (e.g., 'What are the rules regarding rapid cash transfers over \$50k?').

Step 2: Query Embedding

The query is sent to the local Ollama instance running the 'nomic-embed-text' model, converting the text into a numerical vector.

Step 3: Vector Search

The LangChain Chroma client connects to the ChromaDB HTTP server (port 8001). It performs a k-Nearest Neighbors (k-NN) similarity search against the 'vigilos_compliance_rules' collection.

Step 4: Context Injection

The top K most relevant text chunks are extracted, scored, and injected directly into the prompt context for the LLM to make a final compliance judgment.

4. Extraction Code Snippet

```
embeddings = OllamaEmbeddings(model="nomic-embed-text")
client = chromadb.HttpClient(host="localhost", port=8001)
```

```
vectorstore = Chroma(client=client, collection_name="vigilos_compliance_rules",  
embedding_function=embeddings)  
  
# Execute Extraction Pipeline  
results = vectorstore.similarity_search_with_score(query_text, k=3)  
for doc, score in results:  
    print(doc.page_content)
```